

Chapter 5: Crease Density and Decision Boundary Complexity (Experiment 2)

A Technical Report from the Puno Calculus Research Program

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Question

Does crease density predict decision boundary complexity better than layer count or parameter count?

The Puno Calculus predicts that networks with higher crease density (more units operating near their switching threshold) should develop more fragmented decision boundaries, because each near-crease unit defines a boundary that the network is actively folding.

Setup

- Problem:** Multi-scale checkerboard classification (coarse outer cells, fine inner cells)
- Five architectures:**

Name	Layers	Parameters
Shallow (1L)	2	770
Medium (2L)	3	16,769
Deep (4L)	5	64,321
Wide-shallow	2	1,538
Narrow-deep	5	6,177

Metrics: Test accuracy, decision boundary complexity (grid-based crossing count), average crease density across training

Results

Architecture	Best Acc	Boundary Crossing	Avg Crease Density	Layers	Params
Shallow (1L)	80.9%	188	5.10	2	770
Medium (2L)	91.7%	3,120	1.75	3	16,769
Deep (4L)	89.6%	4,096	1.55	5	64,321
Wide-shallow	77.7%	182	6.92	2	1,538
Narrow-deep	91.2%	2,872	1.34	5	6,177

Correlations

Pair	Pearson r
Layers vs complexity	+0.968
Crease density vs complexity	-0.766
Params vs complexity	+0.318
Crease density vs accuracy	-0.556

Key Findings

- 1. Layer count predicts boundary complexity with near-perfect correlation** ($r = +0.968$). Each additional ReLU layer adds folding capacity, producing more fragmented decision boundaries. This validates the "fold depth" concept: deeper networks fold more.
- 2. Crease density is negatively correlated with complexity** ($r = -0.766$). Good networks push units away from creases. A network with high crease density has many "undecided" units that haven't settled into on/off states.
- 3. Crease density drops during training** for all architectures. The RATE of this drop predicts learning quality. Networks that rapidly reduce crease density learn better.
- 4. The wide-shallow network is the worst** — highest crease density (6.92), worst accuracy (77.7%). Width without depth produces many undecided units.
- 5. The narrow-deep network is nearly the best** — lowest crease density (1.34), near-best accuracy (91.2%). Depth allows folding without leaving units at boundaries.

Interpretation

The crease is where learning happens. A unit at its switching threshold (pre-activation near zero) is maximally uncertain — it could go either way. Good training resolves this uncertainty by pushing units away from creases into stable states (either firmly on or firmly off).

This gives us a clear picture of training dynamics from the Puno Calculus perspective: **1. Initialization:** Units are randomly distributed around the crease (high crease density) **2. Early training:** Units quickly settle into on/off states (sharp drop in crease density) **3. Late training:** The partition refines gradually (slow decline in crease density) **4. Convergence:** Most units are far from creases, and the network has decided on a folding pattern